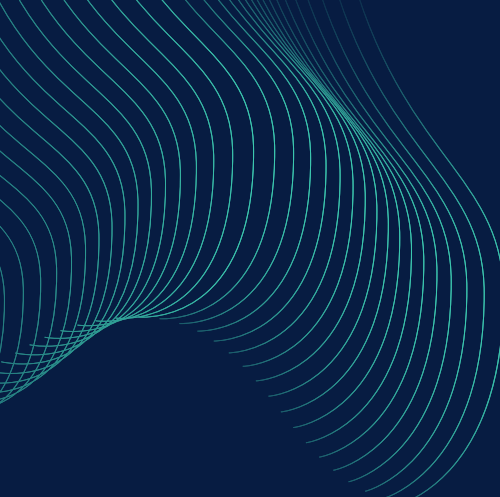
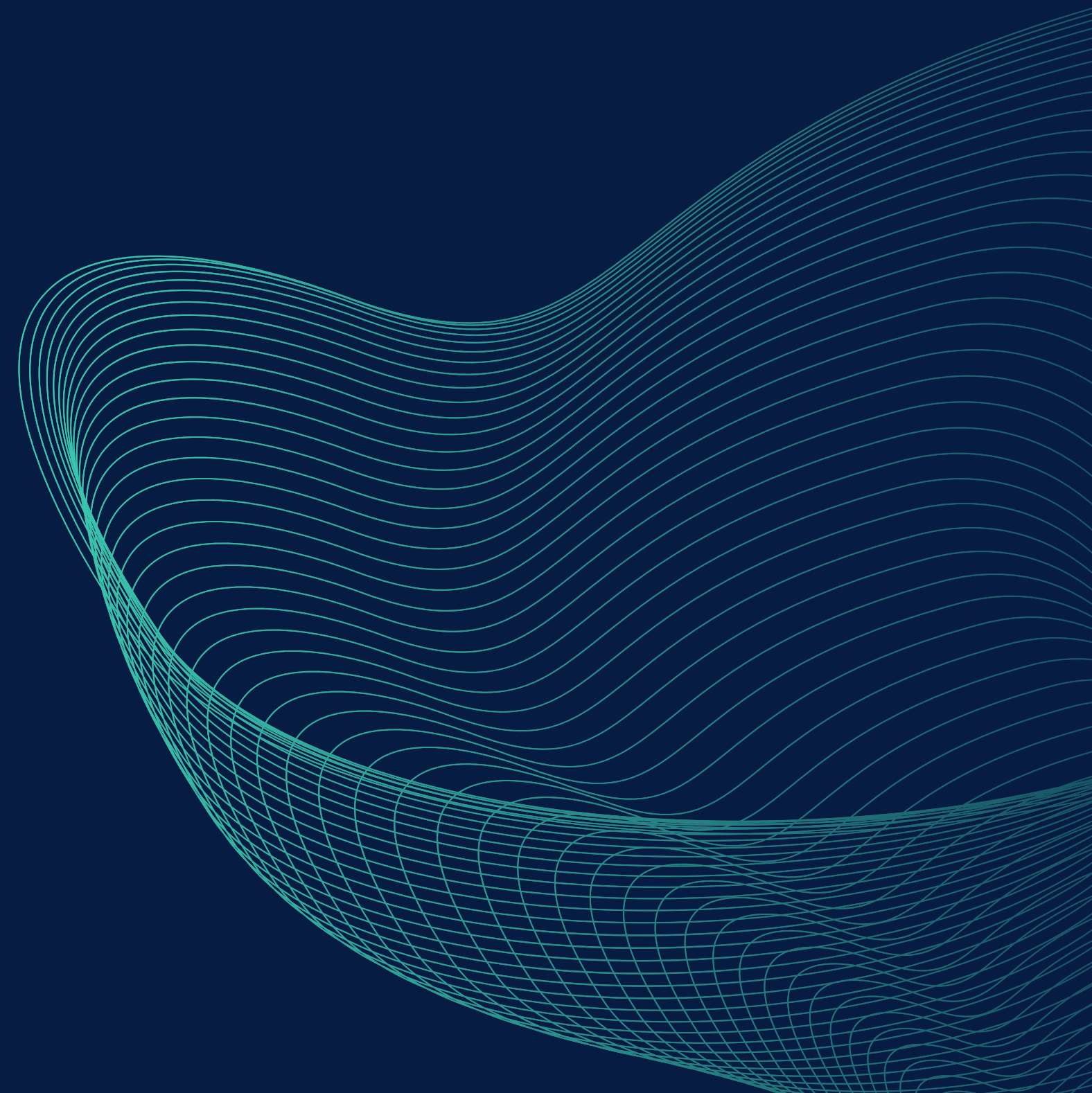




PlayerData

Technical Task

Imogen Sole



Agenda

01

**Approach to the
project**

02

**Execution &
Code**

03

Demonstration

04

**Further Metrics
& Conclusions**

Approach to the Project

Key Considerations:

- **Efficiency**
- **Accuracy and performance**
- **Code structure and understandability**

1. Process the data

- **Remove observations outside the given pitch boundaries**
- **Separate data for players and ball**
- **Remove noise from the speed data**

2. Calculate Metrics

- **Top Speed**
- **Total distance**
- **Total distance in zone 5**
- **Other useful metrics**

3. Create Leaderboards

- **Allow the user to select one of the evaluated metrics**
- **Sort the players based on this metric**
- **Create a leaderboard**

Data Preprocessing

Goals:

- Remove observations outside the given pitch boundaries
- Separate data for players and ball
- Smooth the speed data to remove noise

```
# Read in the data
data = pd.read_csv("data/match_data.csv")

# Filter and sort the data
data_filtered = data[(data['Pitch_x'] >= -52.5) & (data['Pitch_x'] <= 52.5)
                    & (data['Pitch_y'] >= -34) & (data['Pitch_y'] <= 34)]
data_sorted = data_filtered.sort_values(by='Time (s)')

# Separate data for players and ball
all_players_data = data_sorted[data_sorted['participation_id'] != 'ball']
ball_data = data_sorted[data_sorted['participation_id'] == 'ball']

# function to smooth noise in speed measurements
def smooth_speed(player_data):
    player_data['Speed (m/s)'] = player_data['Speed (m/s)'].rolling(window=3).mean()
    return player_data

# Applying the function grouped by 'participation_id'
smooth_speeds = all_players_data.groupby('participation_id').apply(smooth_speed)
all_players_data = smooth_speeds.reset_index(level=0, drop=True)
ball_data = smooth_speed(ball_data)
```

Metrics:

- **Top Speed**

- The highest speed recorded for each player.

- **Total Distance**

- **Total Distance in Speed Zone 5**

- The distance travelled when the speed is in the range 19.8km/h to 25.1km/h.

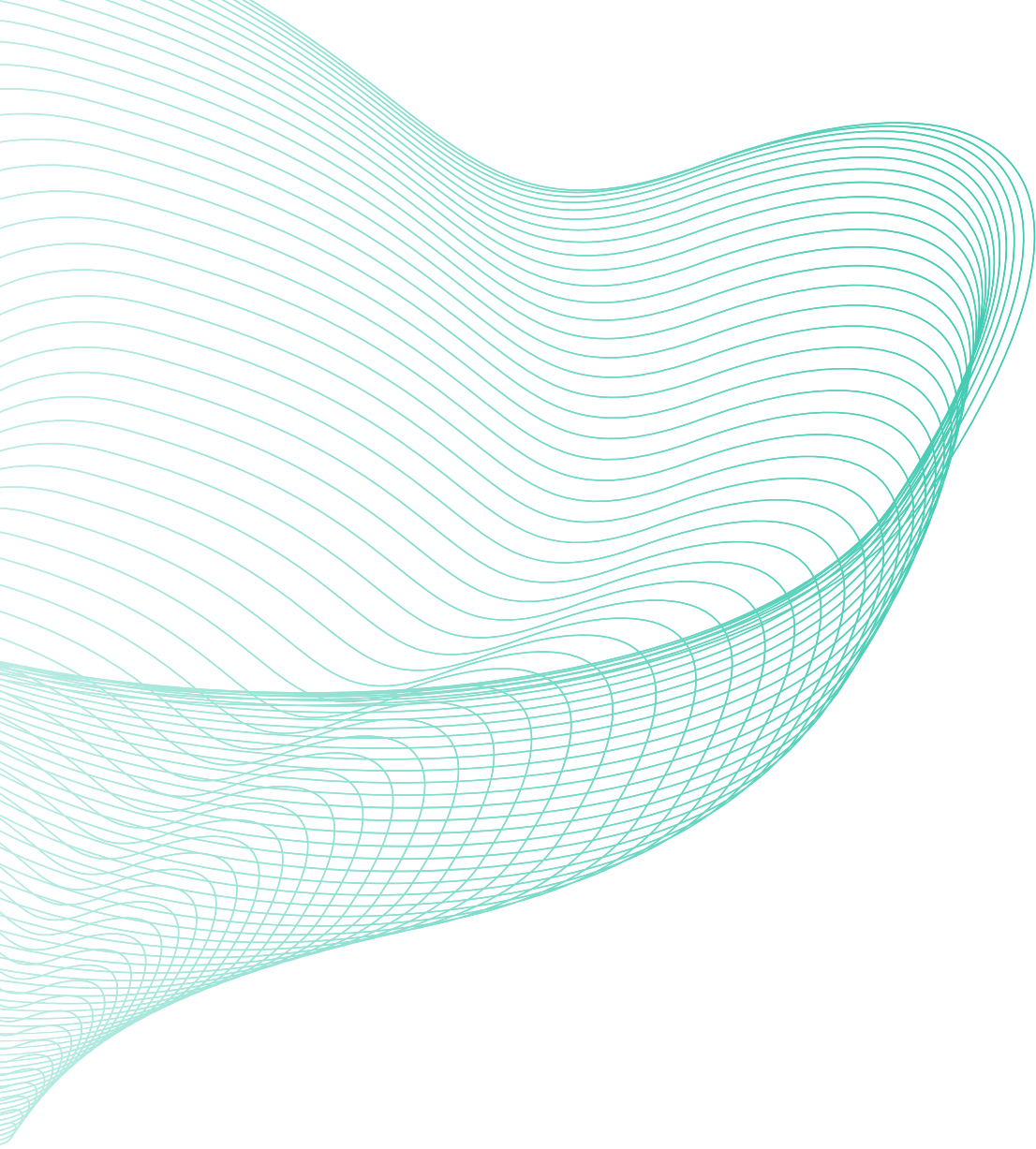
- **Number of Sprint Events**

- Count of the number of times the athlete remained in speed zone 5 for longer than 2 seconds.

- **Time with the ball**

- Duration of time the player was less than 2m away from the ball.

Demonstration



Further Metrics

Performance Metrics

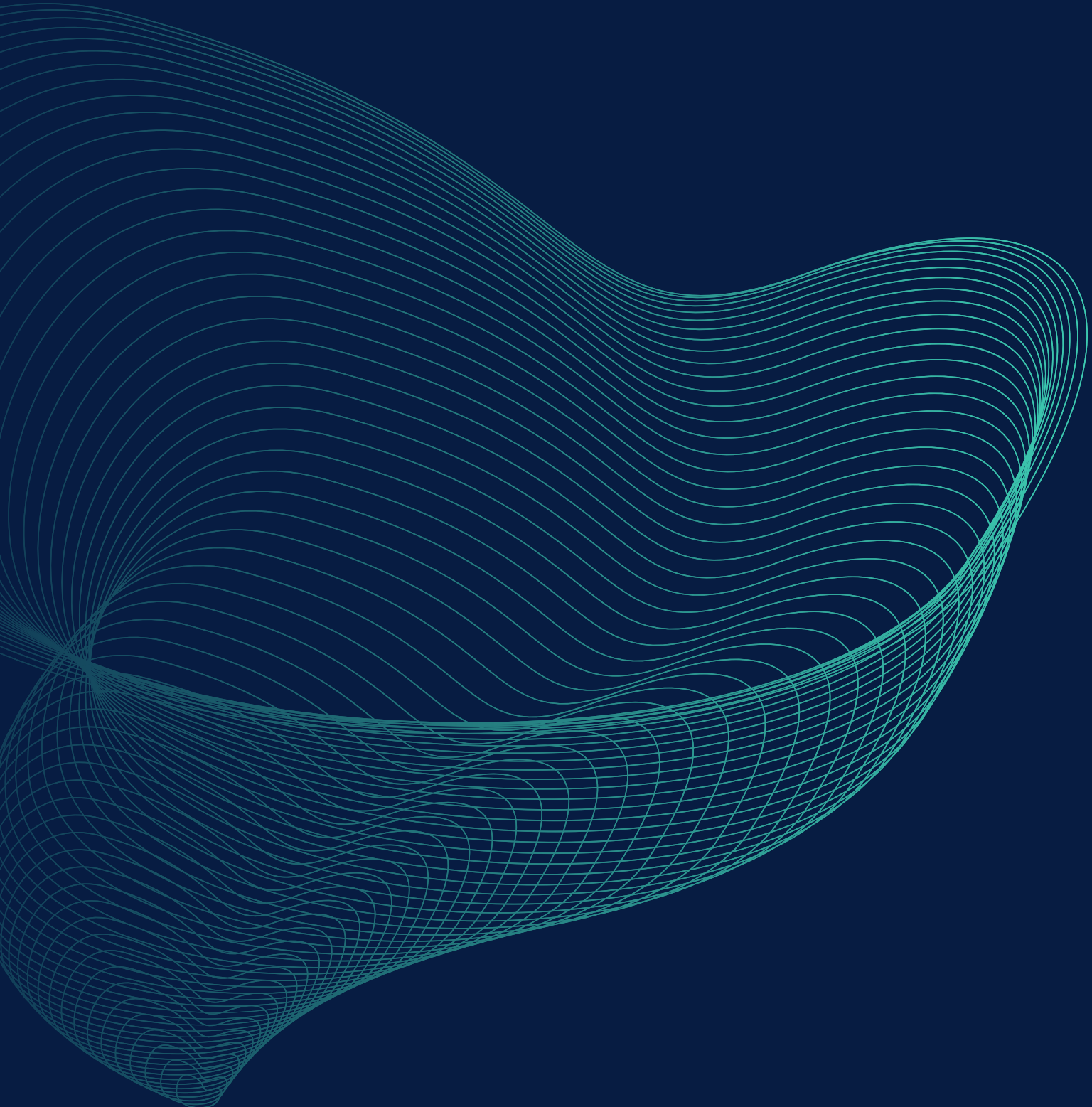
- Pass completion rates
- Intensity of workout
- Number of sprints
- Impacts and collisions
- Acceleration and deceleration
- Top speed on the ball

Injury Prediction

- Using workload and physical impact data to evaluate the risk of injury (1)
- Head Impacts (3)

Helping Referees (VAR decisions)

- Offside rule (2)
- Ambiguous goals



References

1. Cheng, R., and Bergmann, H.M. (2022) 'Impact and workload are dominating on-field data monitoring techniques to track health and well being of team-sports athletes. Available at: [Link](#)
 2. Henderson, A., Lai, D., and Allen, T. (2014) 'A modern Approach to Determine the Offside Law in International Football', Procedia Engineering, Volume 72. Available at: [Link](#)
 3. Patton, D., et.al. (2020) 'Head Impact Sensor Studies in Sports: A Systematic Review of Exposure Confirmation Methods'. Available at: [Link](#)
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